

Eigenvalues & eigenvectors

Observation

$$A \in M_{n \times n}(F)$$

Then A is invertible

$$(\Leftrightarrow) \det(A) \neq 0$$

Proposition:

$$A \in M_{n \times n}(F)$$

TF AE :

(1) A is invertible

$$(2) T: F^n \longrightarrow F^n$$

$$v \longmapsto Av$$

is an isomorphism

rank
- nullity

(3)

T is injective

(4)

T is surjective

(5)

$$\text{RREF}(A) = I_n$$

(6)

$$\det(A) \neq 0$$

Defⁿ:

$$T: V \rightarrow V$$

linear transformation

(1)

$\lambda \in F$ is an eigenvalue

for T if $\exists v \neq 0, v \in V$

s.t. $T(v) = \lambda v$

(2)

In this case, v is an eigenvector for T with

Eigenvalue λ .

Observation 2

$$A \in M_{n \times n}(F)$$

$$T: F^n \longrightarrow F^n$$
$$v \longmapsto Av$$

TFAE

(1) λ is an eigenvalue for T
($\Leftrightarrow$ eigenvalue for A)

(2) $T - \lambda \cdot \text{id}_{F^n}: F^n \longrightarrow F^n$
is not injective

(3) $\text{REF}(A - \lambda I_n)$ has
a free variable

(4)

$$\det(A - \lambda I_n) = 0$$

Observation (3)

$$A = \begin{bmatrix} a_{1,1} & - & - & a_{1,n} \\ a_{2,1} & - & - & a_{2,n} \\ \vdots & & & \vdots \\ a_{n,1} & - & - & a_{n,n} \end{bmatrix}$$

$$A - \lambda I_n = \begin{bmatrix} a_{1,1} - \lambda & - & - & a_{1,n} \\ a_{2,1} & a_{2,2} - \lambda & & a_{2,n} \\ \vdots & & \ddots & \vdots \\ a_{n,1} & a_{n,2} & & a_{n,n} - \lambda \end{bmatrix}$$

Then $\det(A - \lambda I_n)$ is a
polynomial in λ of deg n
and in fact

$$\det(A - \lambda I_n) = (-1)^n \lambda^n - \underbrace{\text{Tr}(A)}_{\text{sum of the diagonal entries}} \lambda^{n-1} + \dots + \det(A)$$

2x2

$$\det \begin{bmatrix} a_{11} - \lambda & a_{12} \\ a_{21} & a_{22} - \lambda \end{bmatrix}$$

$$= (a_{11} - \lambda)(a_{22} - \lambda) - a_{21}a_{12}$$

$$= \lambda^2 - \underbrace{(a_{11} + a_{22})}_{\text{Tr}(A)} \lambda + \underbrace{(a_{11}a_{22} - a_{21}a_{12})}_{\det(A)}$$

3x3

$$\left. \begin{array}{c} (a_{11} - \lambda) \quad \left| \begin{array}{cc} a_{22} - \lambda & a_{23} \\ a_{32} & a_{33} - \lambda \end{array} \right| \end{array} \right\} \begin{array}{l} \text{deg } 3 \\ \text{as a poly} \\ \text{in } \lambda \end{array}$$

$$-a_{2,1} \begin{vmatrix} a_{1,2} & a_{1,3} \\ a_{3,2} & a_{3,3}-\lambda \end{vmatrix} \quad \left. \vphantom{\begin{vmatrix} a_{1,2} & a_{1,3} \\ a_{3,2} & a_{3,3}-\lambda \end{vmatrix}} \right\} \text{deg } 1$$

$$+a_{3,1} \begin{vmatrix} a_{1,2} & a_{1,3} \\ a_{2,2}-\lambda & a_{2,3} \end{vmatrix} \quad \left. \vphantom{\begin{vmatrix} a_{1,2} & a_{1,3} \\ a_{2,2}-\lambda & a_{2,3} \end{vmatrix}} \right\} \text{deg } 1$$

Defn (3) The polynomial

$$\det(A - x I_n) = (-1)^n x^n - \text{Tr}(A)x^{n-1} + \dots + \det(A)$$

is the characteristic polynomial

of A . Denoted $\text{ch}_A(x)$

Observation 2

$$A \in M_{n \times n}(F), \lambda \in F$$

$$T: F^n \longrightarrow F^n$$
$$v \longmapsto Av$$

TFAE

(1) λ is an eigenvalue for T
($\Leftrightarrow$ eigenvalue for A)

(2) $T - \lambda \cdot \text{id}_{F^n}: F^n \longrightarrow F^n$
is not injective

(3) $\text{REF}(A - \lambda I_n)$ has
a free variable

(4) $\det(A - \lambda I_n) = 0$

(5) λ is a zero
for $\chi_A(x)$

Example

$$\bullet F = \mathbb{Z}/5\mathbb{Z}$$

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} \rightarrow \det(A - xI_2)$$

$$x^2 + 1 = \begin{vmatrix} -x & -1 \\ 1 & -x \end{vmatrix}$$

Eigenvalues are $2, 3 \in \mathbb{Z}/5\mathbb{Z}$

• $F = \mathbb{Z}/3\mathbb{Z}$

$$A = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$$

$$\chi_A(x) = x^2 + 1$$

No eigenvalues in $\mathbb{Z}/3\mathbb{Z}$

• $F = \mathbb{Q}, \mathbb{R}$

No eigenvalues in F

• $F = \mathbb{C}$

eigenvalues: $\pm i$

Observation (4)

$$A \in M_{n \times n}(F)$$

Then A has at most n eigenvalues in F

Observation (5)

$$A = \begin{bmatrix} \lambda_1 & & & \\ & \lambda_2 & & \\ & & \ddots & \\ 0 & & & \lambda_n \end{bmatrix}$$

is upper triangular

Then

$$\chi_A(x) = \begin{vmatrix} \lambda_1 - x & & & * \\ & \lambda_2 - x & & \\ & & \ddots & \\ & & & \lambda_n - x \end{vmatrix}$$

$$= (\lambda_1 - x)(\lambda_2 - x) \cdots (\lambda_n - x)$$

$\Rightarrow$ Eigenvalues of $A = \{\lambda_1, \lambda_2, \dots, \lambda_n\}$